1. 通电电流的磁场 B 计算公式

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I\,d\vec{l} \times \hat{r}}{r^2}$$ (magnetic field of a current elem

2. 运动电荷的磁场 B 计算公式

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{q\vec{v} \times \hat{r}}{r^2}$$ (magnetic field of a poin with constant velocity)

3. 求感生电动势 emf 表达式是

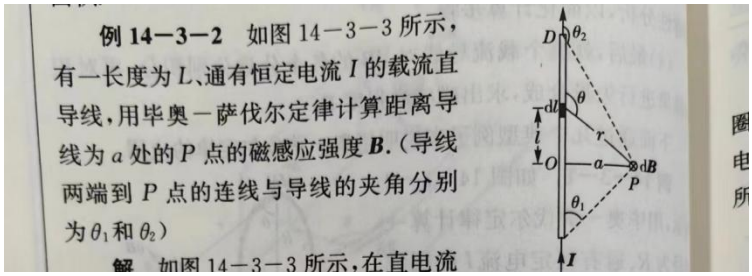
$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$ (stationary integration path)

4. 求无限长的长直螺线管内部的磁感应强度 B

$$B = \frac{\mu_0 I}{2\pi r}$$

5. 下图有限长的导线在 P 点产生的磁场 B 是多少？



例 14—3—2 如图 14—3—3 所示，有一长度为 L、通有恒定电流 I 的载流直导线，用毕奥—萨伐尔定律计算距离导线为 a 处的 P 点的磁感应强度 $\boldsymbol{B}$.（导线两端到 P 点的连线与导线的夹角分别为 θ_1 和 θ_2）

解 如图 14—3—3 所示，在直电流

(4) $x \gg R$ 时，$B=\frac{\mu_0 IR^2}{2x^3}$，$B=\frac{\mu_0 IS}{2\pi x^3}$，式中 $S=\pi R^2$ 为圆电流面积.

例 14—3—2 如图 14—3—3 所示，有一长度为 L、通有恒定电流 I 的载流直导线，用毕奥—萨伐尔定律计算距离导线为 a 处的 P 点的磁感应强度 $\boldsymbol{B}$.（导线两端到 P 点的连线与导线的夹角分别为 θ_1 和 θ_2）

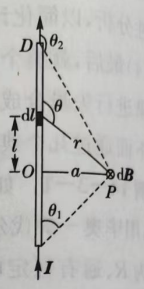


图 14—3—3 有限长载流直导线的磁场

解 如图 14—3—3 所示，在直电流上取电流元 $I\mathrm{d}\boldsymbol{l}$，由毕奥—萨伐尔定律知电流元 $I\mathrm{d}\boldsymbol{l}$ 在任意一点 P 产生的磁感应强度 $\mathrm{d}\boldsymbol{B}$ 的大小为

$$\mathrm{d}B=\frac{\mu_0 I\mathrm{d}l\sin\theta}{4\pi r^2}$$

方向垂直纸面向内，各电流元在点 P 产生的磁感应强度 $\mathrm{d}\boldsymbol{B}$ 方向相同.

统一将上式中变量用 θ 表示，则

$$l=-a\cot\theta,\ \mathrm{d}l=\frac{a\mathrm{d}\theta}{\sin^2\theta},\ r=\frac{a}{\sin\theta}$$

对 $\mathrm{d}B$ 积分，可得 $\boldsymbol{B}$ 的大小为

$$B=\frac{\mu_0 I}{4\pi a}\int_{\theta_1}^{\theta_2}\sin\theta\mathrm{d}\theta=\frac{\mu_0 I}{4\pi a}(\cos\theta_1-\cos\theta_2)$$

$\boldsymbol{B}$ 的方向与直电流成右手螺旋关系，四指的绕向即为 $\boldsymbol{B}$ 的方向.

讨论

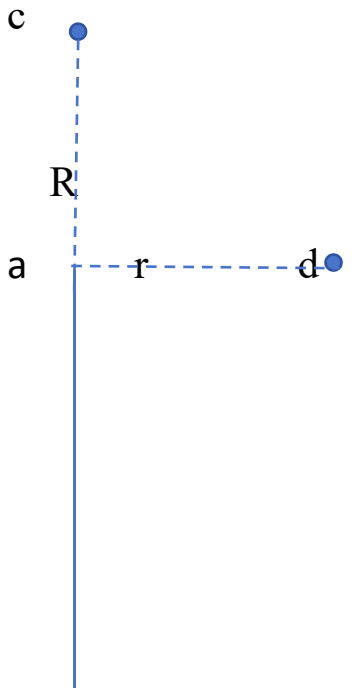
(1) 载流直导线为“无限长”时，$\theta_1=0$，$\theta_2=\pi$，$B=\frac{\mu_0 I}{2\pi a}$；

(2) 载流直导线为“半无限长”时，$\theta_1=\pi/2$，$\theta_2=\pi$，$B=\frac{\mu_0 I}{4\pi a}$.

例 14—3—3 直螺线管是指长

6. 特殊情况下的磁场

（1）半无穷长的导线 a 点固定，b 点在无穷远



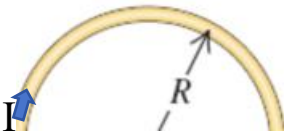
b

求 c，d 点磁感应强度 B 分别是

c 点的 B 为 0

d 点的 B 为 $\frac{\mu_0 I}{4\pi R}$

(2) I



O

求 O 点磁感应强度 B

$$\frac{\mu_0 I}{4a}$$

7. 磁场的来源

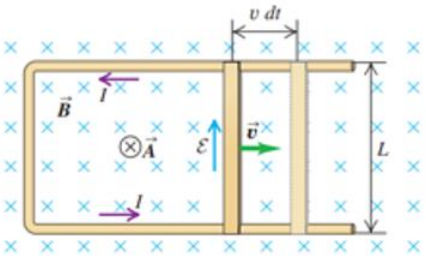
磁铁

运动电荷

电流

变化的电场

8. 一根电线切割磁感应线产生动生电动势的公式，如何判断方向。



$$\mathcal{E} = \oint (\vec{v} \times \vec{B}) \cdot d\vec{l} = -BLv$$

判断电动势的方向，可以用 $\vec{v} \times \vec{B}$ 的方向，也就是在感应电场下正电荷的运动方向

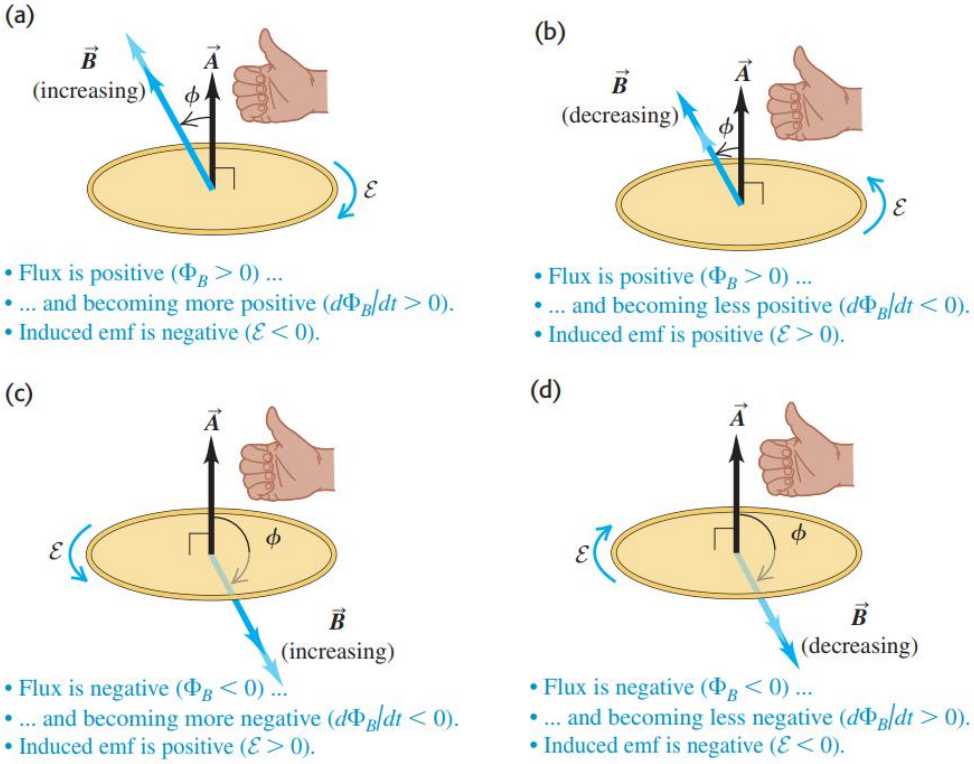
9. B 和 A 夹角为θ，θ发生变化，对于的感生电动势 emf 等于多少

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

其中Φ_B=BAcosθ，代入

10. A 已知，磁感应强度 B 变化，感生电动势 emf 等于多少 产生感生电动势的方向判断。

6 The magnetic flux is becoming (a) more positive, (b) less positive, (c) more negative, and (d) less negative. Therefore Φ_B is increasing in (a) and (d) and decreasing in (b) and (c). In (a) and (d) the emfs are negative (they are opposite to the direction of the curled fingers of your right hand when your right thumb points along $\vec{A}$). In (b) and (c) the emfs are positive (in the same direction as the curled fingers).



11. 在圆形电容器内部的位移电流表达式？ 在电容器内部距离中轴半径为 r 的位置处的磁场是多少？dq/dt 充电速度和传导电流的关系？

21.7 Displacement Current and Maxwell's Equations

- Generalizing Ampere's Law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}}$$

$$q = Cv = \frac{\epsilon A}{d}(Ed) = \epsilon E A = \epsilon \Phi_E$$

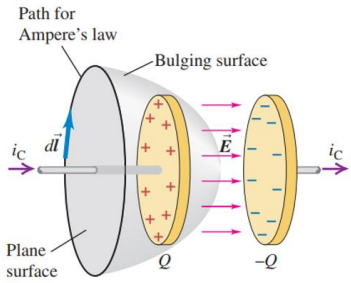
$$i_C = \frac{dq}{dt} = \epsilon \frac{d\Phi_E}{dt}$$

$$i_D = \epsilon \frac{d\Phi_E}{dt} \qquad \text{(displacement current)}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (i_C + i_D)_{\text{encl}} \qquad \text{(generalized Ampere's law)}$$

$$j_D = \epsilon \frac{dE}{dt}$$

21 Parallel-plate capacitor being charged. The conduction current through the plane surface is i_C, but there is no conduction current through the surface that bulges out to pass between the plates. The two surfaces have a common boundary, so this difference in I_{encl} leads to an apparent contradiction in applying Ampere's law.



$$i_D = \epsilon \frac{d\Phi_E}{dt} \qquad \text{(displacement current)}$$

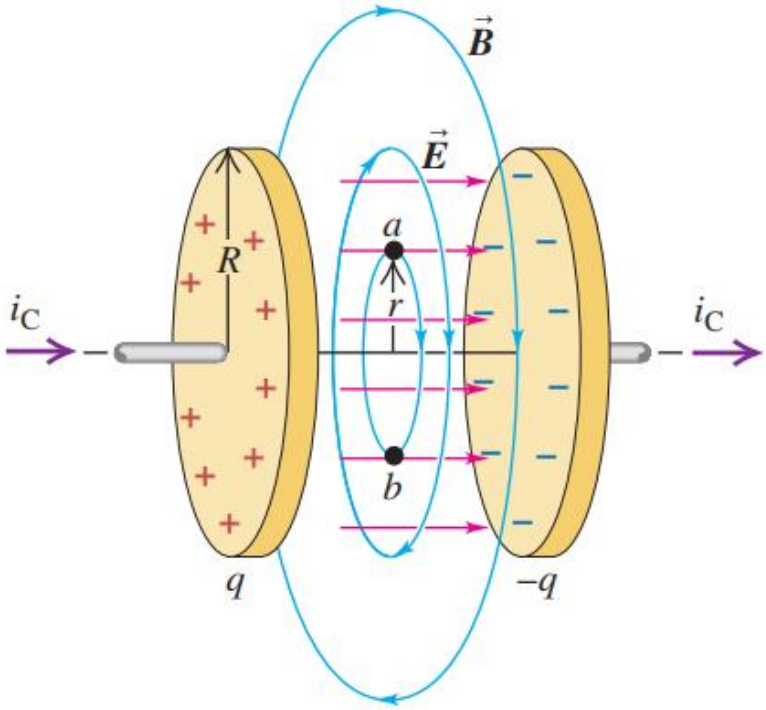
$$i_C = \frac{dq}{dt} = \epsilon \frac{d\Phi_E}{dt}$$

总的位移电流等于总的传导电流，但是面积不一样，传导电流的面积是导线，位移电流的总面积对应的是电容器截面、

$$\oint \vec{B} \cdot d\vec{l} = 2\pi r B = \mu_0 \frac{r^2}{R^2} i_C \qquad \text{or}$$

$$B = \frac{\mu_0}{2\pi} \frac{r}{R^2} i_C$$

22 A capacitor being charged by a current i_C has a displacement current equal to i_C between the plates, with displacement-current density $j_D = \epsilon\, dE/dt$. This can be regarded as the source of the magnetic field between the plates.

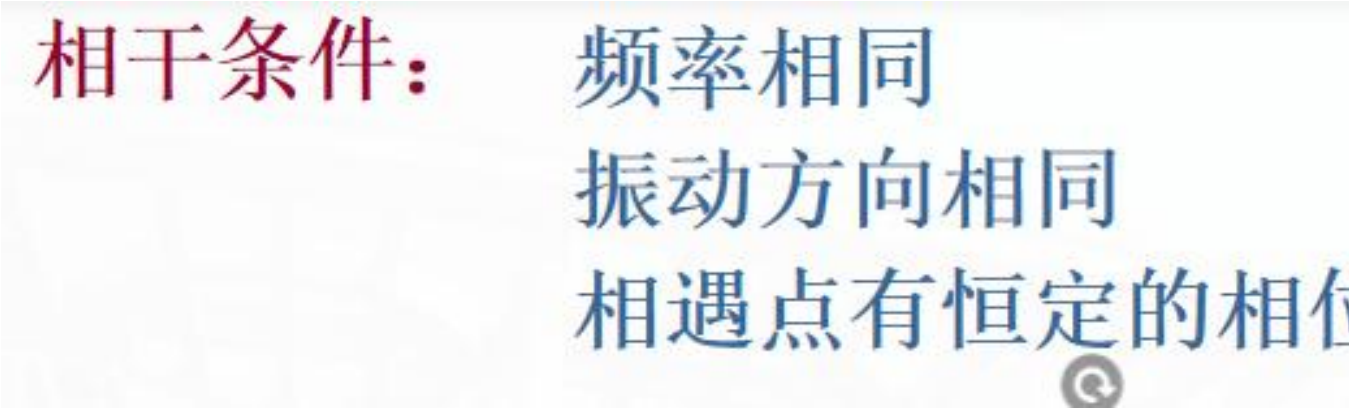


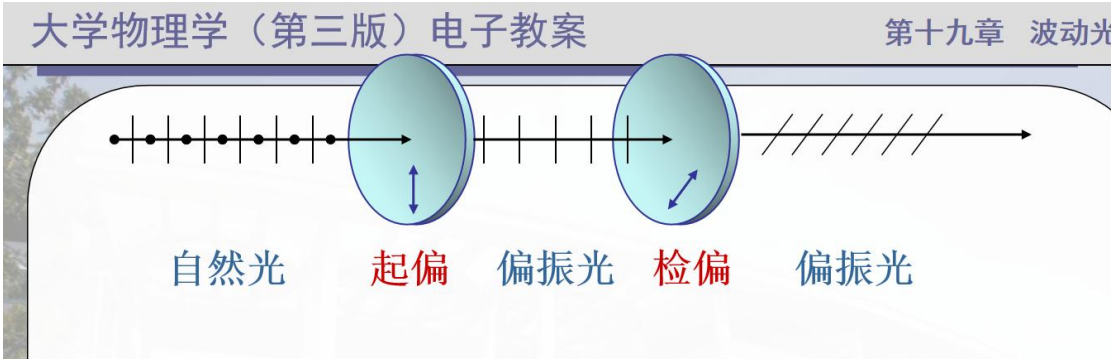
12. 电流产生磁场，变化的电流产生变化的磁场，变化的磁场又会产生感应电场。

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i_C + \epsilon_0 \frac{d\Phi_E}{dt} \right)_{\text{encl}} \qquad \text{(Ampere's law)}$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} \qquad \text{(Faraday's law)}$$

13. 两列波的相干条件



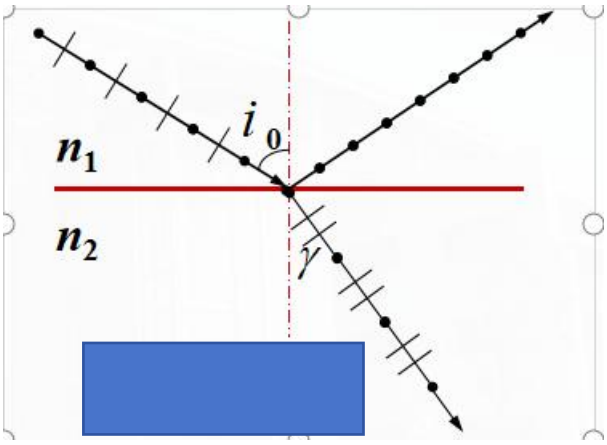
14. 

I_0 I I’

求 I 和 I’

$I=I_0/2$

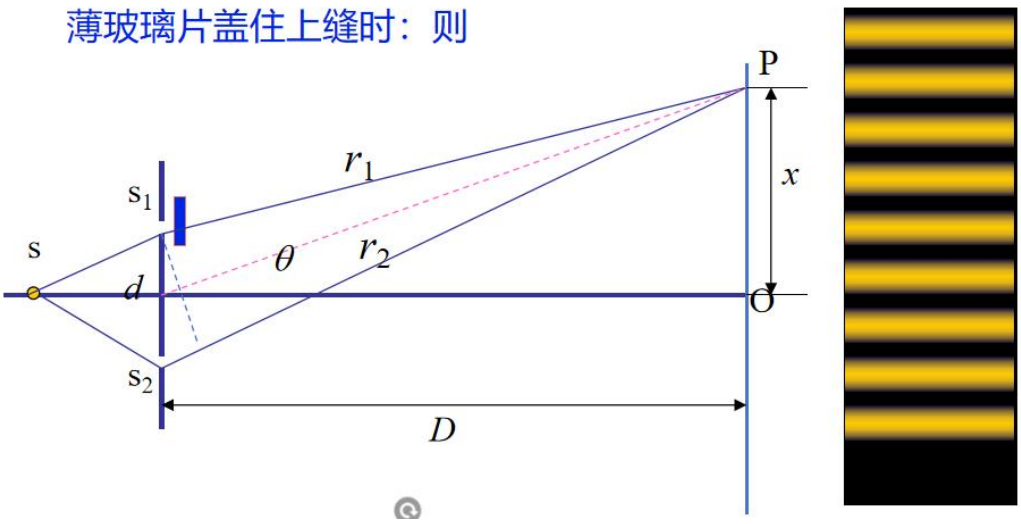
$I'=I(\cos\theta)^2=(I_0/2)(\cos\theta)^2$

15. 

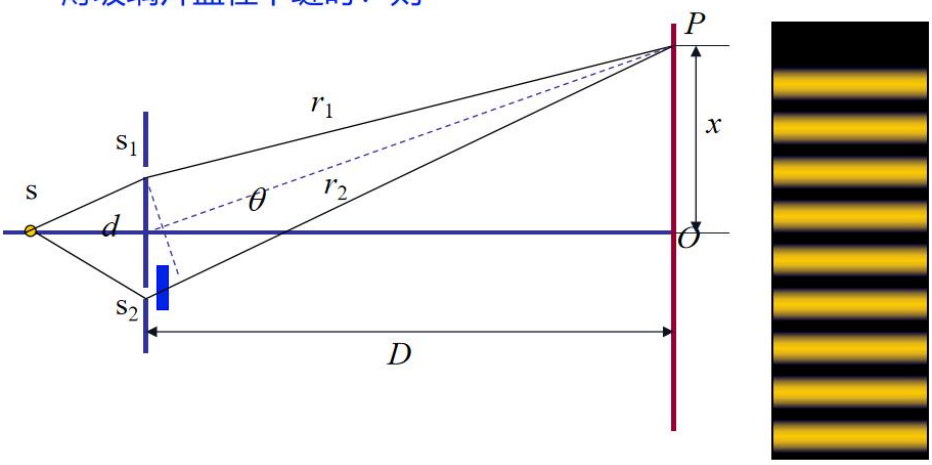
给出图中四个参数的关系

$n_1 \sin i_0 = n_2 \sin \gamma$

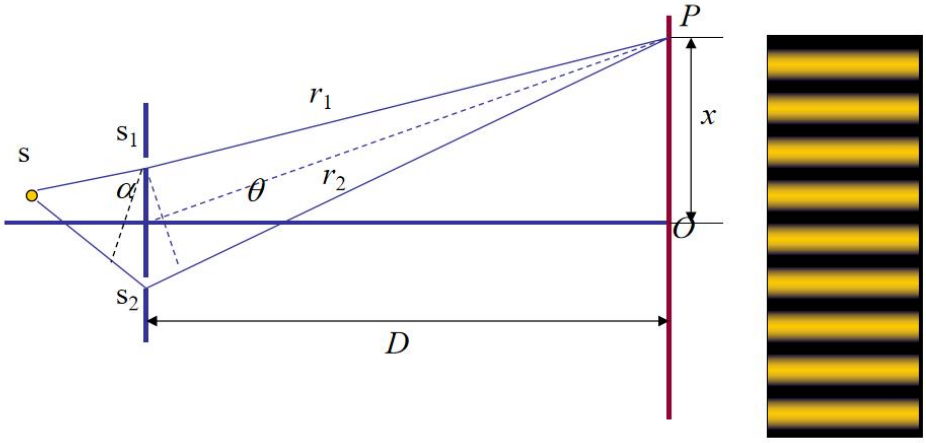
16. 下面中央明纹移动情况



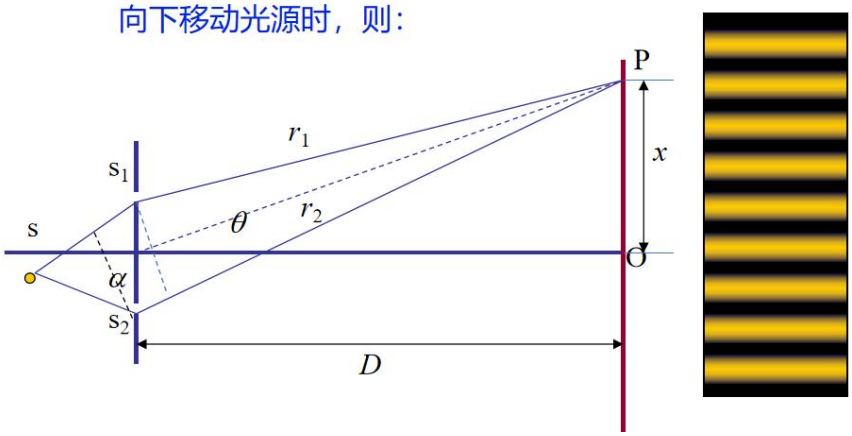
薄玻璃片盖住下缝时：则



向上移动光源时，则：



向下移动光源时，则：

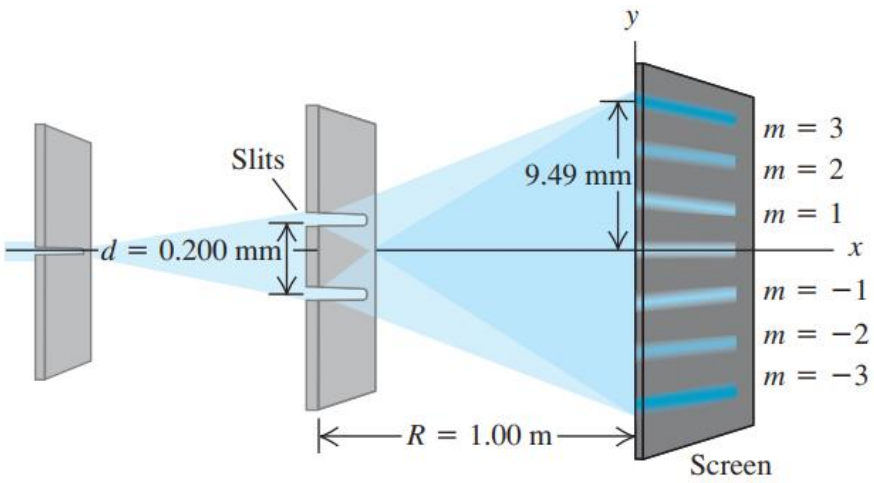


参照 ppt

17.

Figure 7 shows a two-slit interference experiment in which the slits are 0.200 mm apart and the screen is 1.00 m from the slits. The $m = 3$ bright fringe in the figure is 9.49 mm from the central fringe. Find the wavelength of the light.

7 Using a two-slit interference experiment to measure the wavelength of light.



明纹的坐标 y_m 是多少。

相邻明纹间距是多少。

双缝

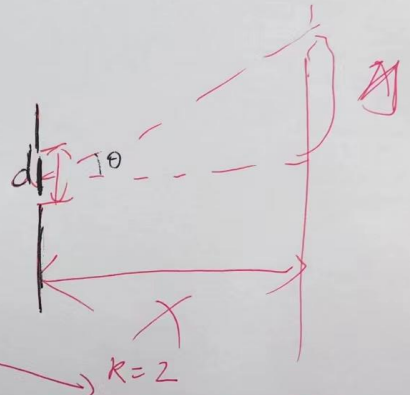
明纹

双缝间距

$$d\sin\theta = \pm k\lambda$$

$$\tan\theta = \frac{y}{x}$$

$k = 0, 1, 2, 3, \ldots$



k=1

一级明纹

$$d\sin\theta_1 = \lambda \quad ①$$

$$\tan\theta_1 = \frac{y_1}{x} \quad ②$$

θ很小 $\sin\theta_1 \approx \tan\theta_1$

二级明纹

$$d\sin\theta_2 = 2\lambda \quad ③$$

$$\tan\theta_2 = \frac{y_2}{x} \quad ④$$

θ很小 $\sin\theta_2 \approx \tan\theta_2$

① ÷ ②

$$d = \frac{\lambda x}{y_1} \quad ⑤$$

③ ÷ ④

$$d = \frac{2\lambda x}{y_2} \quad ⑥$$

$$\Delta y = y_2 - y_1 = \frac{2\lambda x}{d} - \frac{\lambda x}{d} = \frac{\lambda x}{d}$$

Δy 是一级明纹与二级明纹的间距.

同理也是二级明纹与三明纹的间距

也是 n 级明纹与 n+1 级明纹的间距.

也是 n 级暗纹与 n+1 级暗纹的间距

暗纹

$d\sin\theta = \pm(2k+1)\frac{\lambda}{2}$

$\tan\theta = \frac{y}{x}$

$k = 0, 1, 2, 3 \ldots$

$\downarrow k=0$

一级暗纹

$d\sin\theta_{1D} = \frac{\lambda}{2}$ ①

$\tan\theta_{1D} = \frac{y_{1D}}{x}$ ②

二级暗纹

$d\sin\theta_{2D} = \frac{3\lambda}{2}$ ③

$\tan\theta_{2D} = \frac{y_{2D}}{x}$ ④

①/②

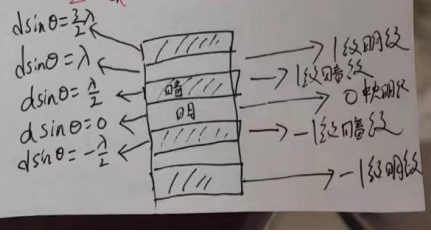
$$d = \frac{\lambda x}{2y_{1D}} \quad ⑤$$

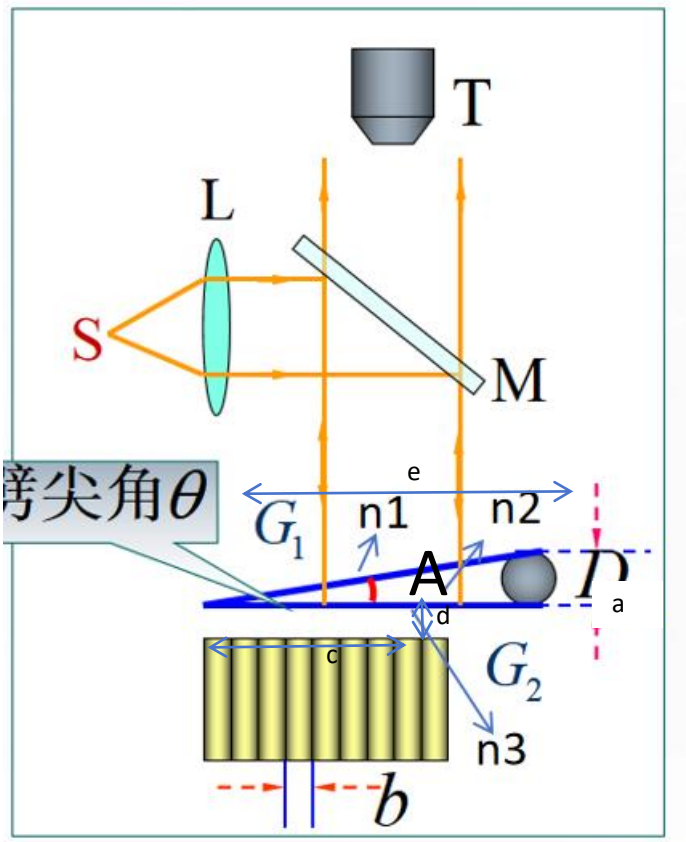
③/④

$$d = \frac{3\lambda x}{2y_{2D}} \quad ⑥$$

$$\Delta y' = y_{2D} - y_{1D} = \frac{3\lambda x}{2d} - \frac{\lambda x}{2d} = \frac{\lambda x}{d}$$

纹宽度一样



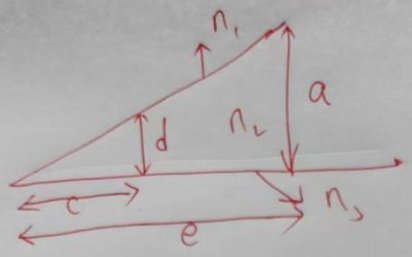
18. 

上图 n1 和 n3 是同一种东西，（1）要不要考虑半波损失，（2）A 点的光程差，（3）明纹条件和暗纹条件，（4）a,d,c,e 之间的关系相似三角形（a/e=d/c）。暗纹所在的位置对应的 c 等于多少。（5）两个面的交线是明纹还是暗纹。（6）b 等于多少。

$n_1 = n_3$

劈尖 n_2空气

$\therefore n_2 = 1$



$$\frac{d}{c} = \frac{a}{e}$$

(1) 夏. 汉堡结构

(两个相同的夫位另一个的情况) 考虑半波损失

(2) $\Delta = 2nd + \frac{\lambda}{2}$

(3) $\Delta = 2n_2 d + \frac{\lambda}{2} = \begin{cases} k\lambda & \text{明} \\ (2k+1)\frac{\lambda}{2} & \text{暗} \end{cases}$

$d > 0$

$\therefore \Delta > 0$

$\therefore \begin{cases} k\lambda & k = 1, 2 \cdots \text{明} \\ (2k+1)\frac{\lambda}{2} & k = 0, 1, 2 \cdots \text{暗} \end{cases}$

(4) 暗纹

$2n_2 d + \frac{\lambda}{2} = 2k \cdot \frac{\lambda}{2} + \frac{\lambda}{2}$

$2n_2 d = 2k\frac{\lambda}{2}$

$n_2 d = k\frac{\lambda}{2}$ $k =$

$n_2 = 1$

$d = k\frac{\lambda}{2}$

$\frac{d}{c} = \frac{a}{e}$

$c = \frac{de}{a} = \left(k\frac{\lambda}{2}\right)\frac{e}{a}$

$k = 0, 1, 2 \cdots$

(5) 当 $d = 0$ 时

$\Delta = \frac{\lambda}{2}$ 满足暗纹条件

$\therefore$ 是暗纹

(6) b 是相邻条纹间距 $b = \frac{\lambda}{2}\frac{e}{a}$

19.

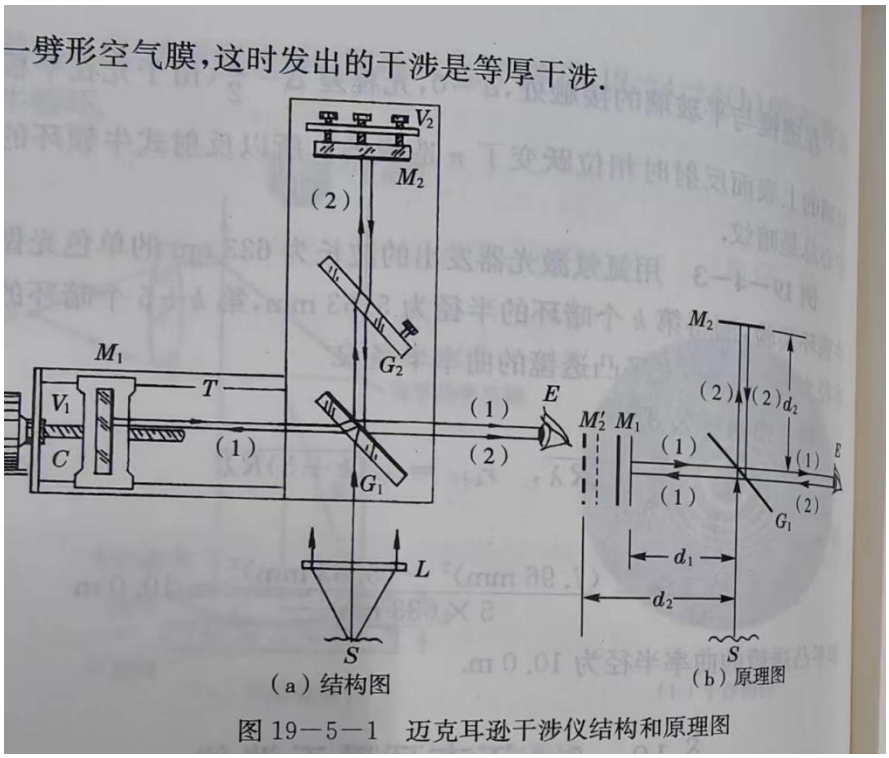


图 19—5—1　迈克耳逊干涉仪结构和原理图

当 M1 移动距离 d 时，光程差变化为 2d(因为是来回)。

如果光程差变化λ，条纹移动一个。所以光程差变化 2d，条纹移动 n 关系如下

$$\frac{n}{2d} = \frac{1}{\lambda}$$

$$n = \frac{2d}{\lambda}$$

20. 一束光通过两种介质 n1 和 n2，在两种介质中的速度是 v1 和 v2，在两种介质中的波长是λ1 和λ2。速度和折射率的关系？波长和折射率的关系？在两种介质中的频率一样吗？

空气折射率 n 等于多少？

频率一样不变

$, n = c/v$ n1=c/v1 n2=c/v2

$, \lambda = \lambda_0/n$ λ1= λ 0/n1 λ2= λ 0/n2

Example 2 **Index of refraction in the eye**

The wavelength of the red light from a helium-neon laser is 633 nm in air but 474 nm in the aqueous humor inside your eyeball. Calculate the index of refraction of the aqueous humor and the speed and frequency of the light in it.

SOLUTION

IDENTIFY and SET UP: The key ideas here are (i) the definition of index of refraction n in terms of the wave speed v in a medium and the speed c in vacuum, and (ii) the relationship between wavelength λ_0 in vacuum and wavelength λ in a medium of index n. We use Eq. (1), $n = c/v$; Eq. (5), $\lambda = \lambda_0/n$; and $v = \lambda f$.

EXECUTE: The index of refraction of air is very close to unity, so we assume that the wavelength λ_0 in vacuum is the same as that in air, 633 nm. Then from Eq. (5),

$$\lambda = \frac{\lambda_0}{n} \qquad n = \frac{\lambda_0}{\lambda} = \frac{633\ \text{nm}}{474\ \text{nm}} = 1.34$$

This is about the same index of refraction as for water. Then, using $n = c/v$ and $v = \lambda f$, we find

$$v = \frac{c}{n} = \frac{3.00 \times 10^8\ \text{m/s}}{1.34} = 2.25 \times 10^8\ \text{m/s}$$

$$f = \frac{v}{\lambda} = \frac{2.25 \times 10^8\ \text{m/s}}{474 \times 10^{-9}\ \text{m}} = 4.74 \times 10^{14}\ \text{Hz}$$

EVALUATE: Note that while the speed and wavelength have different values in air and in the aqueous humor, the *frequency* in air, f_0, is the same as the frequency f in the aqueous humor:

$$f_0 = \frac{c}{\lambda_0} = \frac{3.00 \times 10^8\ \text{m/s}}{633 \times 10^{-9}\ \text{m}} = 4.74 \times 10^{14}\ \text{Hz}$$

When a light wave passes from one material into another, the wave speed and wavelength both change but the wave frequency is unchanged.

21. 单缝衍射

明纹的坐标 ym 是多少。

暗纹的坐标 ym 是多少。

一级暗纹和负一暗纹间距（即中央明纹的宽度）是多少？

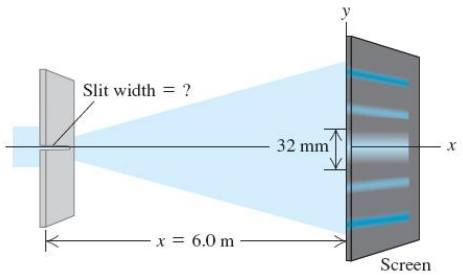
Example 1 Single-slit diffraction

You pass 633-nm laser light through a narrow slit and observe the diffraction pattern on a screen 6.0 m away. The distance on the screen between the centers of the first minima on either side of the central bright fringe is 32 mm (Fig. 7). How wide is the slit?

SOLUTION

IDENTIFY and SET UP: This problem involves the relationship between the positions of dark fringes in a single-slit diffraction

7 A single-slit diffraction experiment.



明条纹

$a\sin\theta = \pm k\lambda$

$k = 1, 2, 3.$

$\tan\theta = \frac{y}{x}$

$k = 1$

一级明纹

$a\sin\theta_{1D} = \lambda$ ①

$\tan\theta_{1D} = \frac{y_{1D}}{x}$ ②

二级明纹

$a\sin\theta_{2D} = 2\lambda$ ③

$\tan\theta_{2D} = \frac{y_{2D}}{x}$ ④

①/②

$a = \frac{\lambda x}{y_{1D}}$

③/④

$a = \frac{2\lambda x}{y_{2D}}$

$$\Delta y' = y_{2D} - y_{1D} = \frac{2\lambda x}{a} - \frac{\lambda x}{a} = \frac{\lambda x}{a}$$

-1级明纹

$a\sin\theta_{-1D} = -\lambda$ ⑤

$\tan\theta_{-1D} = \frac{y_{-1D}}{x}$ ⑥

⑤/⑥得

$a = \frac{-\lambda x}{y_{-1D}}$

$$\Delta y'' = y_{1D} - y_{-1D} = \frac{\lambda x}{a} - \left(-\frac{\lambda x}{a}\right) = \frac{2\lambda x}{a}$$

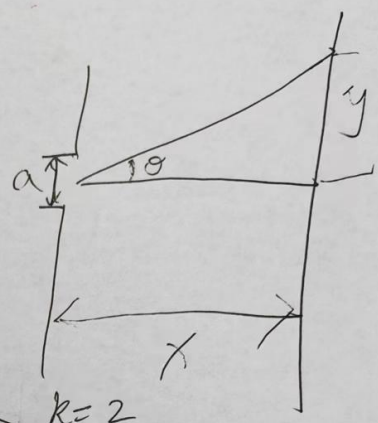
单缝

明纹

$a\sin\theta = 0$ 中央明纹

$a\sin\theta = \pm(2k+1)\frac{\lambda}{2}$ 其它明纹 $k=1,2,3\cdots$

$\tan\theta = \frac{y}{x}$



$k=1$

一级明纹

$a\sin\theta_1 = \frac{3}{2}\lambda$ ①

$\tan\theta_1 = \frac{y_1}{x}$ ②

$k=2$

二级明纹

$a\sin\theta_2 = \frac{5}{2}\lambda$ ③

$\tan\theta_2 = \frac{y_2}{x}$ ④

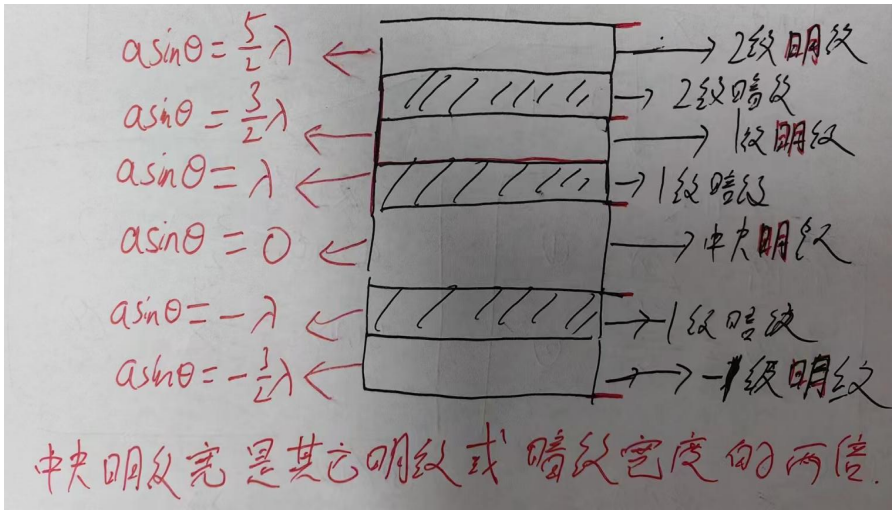
①/②

$$a = \frac{3\lambda x}{2y_1}$$

③/④

$$a = \frac{5\lambda x}{2y_2}$$

$$\Delta y = y_2 - y_1 = \frac{5\lambda x}{2a} - \frac{3\lambda x}{2a} = \frac{\lambda x}{a}$$


$a\sin\theta=\frac{5}{2}\lambda$
$a\sin\theta=\frac{3}{2}\lambda$
$a\sin\theta=\lambda$
$a\sin\theta=0$
$a\sin\theta=-\lambda$
$a\sin\theta=-\frac{3}{2}\lambda$
2级明纹
2级暗纹
1级明纹
1级暗纹
中央明纹
1级暗纹
-1级明纹
中央明纹宽是其它明纹或暗纹宽度的两倍.